DATA SCIENCE SUBJECT DIGITAL LIBRARY ? STUDY GUIDE

Module 4
Statistical Thinking

A practical introduction to statistical reasoning through distributions, variability, probability functions, percentiles, random sampling, and common probability models.

11 teaching periods ? 10 topics ? Learning objectives and full topic notes

Learning Objectives

- Represent data with histograms and summarize distributions using measures such as variance, standard deviation, central tendency, and spread.
- Identify outliers and explain how they may arise from measurement error, unusual observations, or genuine rare events.
- Construct and interpret probability mass functions and understand how they represent discrete random variables.
- Explain the class size paradox and recognize why averages can be misleading when groups have unequal sizes.
- Use DataFrame indexing and selection techniques to isolate and analyze relevant subsets of data.
- Represent cumulative distribution functions and use percentiles to compare observations across different datasets.
- Generate random numbers and understand the difference between random variation and systematic bias.
- Model data using exponential, normal, and lognormal distributions and evaluate whether an observed distribution is consistent with a chosen model.

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Topic Notes

TOPIC 01

Representing and Plotting Histograms

A histogram divides the range of a numerical variable into consecutive, non-overlapping bins and counts how many observations fall in each bin, displaying the count as the height of a bar. Histograms reveal the shape of a distribution: bell-shaped and symmetric (normal), right-skewed (long tail on the right), left-skewed (long tail on the left), uniform, or bimodal (two peaks). In Python, plt.hist(data, bins=20) and sns.histplot(data, bins=20) create histograms. The number of bins is a critical choice â€” rule-of-thumb methods include Sturge's rule (k = 1 + log2n). Normalizing the histogram (density=True) converts counts to probability density, enabling comparison across datasets of different sizes.

TOPIC 02

Outliers and Summarizing Distributions

An outlier is an observation that lies an abnormal distance from other values in the dataset. Outliers can arise from: (1) measurement or recording errors (a typo entering 1000 instead of 100); (2) genuine extreme values (a billionaire in an income survey); or (3) sampling from a different population (equipment malfunction). Detection methods include the IQR rule (flag values below Q1 - 1.5xIQR or above Q3 + 1.5xIQR), z-score thresholds (|z| > 3), and visual inspection via box plots. Summarizing a distribution requires reporting central tendency (mean, median, mode) and spread (range, IQR, variance, standard deviation) together, as a single number never fully characterizes a distribution.

TOPIC 03

Variance, Standard Deviation, and Reporting Statistical Results

Variance measures how spread out values are around the mean: population variance = sum of squared deviations divided by N; sample variance uses (n-1) as the denominator (Bessel's correction) to make it an unbiased estimator. Standard deviation is the square root of variance, expressed in the same units as the original data, making it more interpretable. When reporting statistical results, always accompany a mean with its standard deviation: 'The mean exam score was 72.4 plus or minus 8.3 (SD)'. For skewed distributions, report the median and IQR instead, as they are more robust to outliers.

TOPIC 04

Probability Mass Functions and PMF Plotting

A Probability Mass Function (PMF) assigns a probability to each possible value of a discrete random variable such that all probabilities sum to 1. For example, the PMF of a fair six-sided die assigns probability 1/6 to each value {1, 2, 3, 4, 5, 6}. An empirical PMF is computed from observed data by normalizing the frequency of each distinct value: P(x) = count(x) / n. PMFs are visualized as bar charts where the height of each bar is the probability (not the count). They are particularly useful for comparing two datasets of different sizes on the same scale, and for identifying which values are over- or under-represented compared to a theoretical distribution.

TOPIC 05

Other Statistical Visualizations and DataFrame Indexing

Beyond histograms and PMFs, several other visualizations support statistical analysis. Box plots display the five-number summary (minimum, Q1, median, Q3, maximum) and flag outliers as individual points â€” ideal for comparing distributions across multiple groups. Violin plots combine box plots with KDE curves, showing distributional shape alongside summary statistics. In Python, Pandas DataFrames support powerful indexing: df.loc[row_label, col_label] selects by label, df.iloc[row_int, col_int] selects by integer position, and boolean indexing df[df['column'] > value] filters rows. These techniques enable rapid extraction of subsets for group-specific statistical analysis.

TOPIC 06

The Class Size Paradox and Selection Bias

The class size paradox is a striking example of how averages can mislead when groups have unequal sizes. Suppose a university has 4 small courses with 20 students each and 1 large lecture with 400 students. The average class size from the course perspective is (4x20 + 400) / 5 = 96 students. But the average from the student perspective â€” asking each student how large is their class â€” is much larger, because students are much more likely to be in the large class. This is selection bias: the sample is not representative of the population because the sampling probability is proportional to group size. Recognizing such biases is critical for correct statistical interpretation.

TOPIC 07

Cumulative Distribution Functions and the Limits of PMFs

A Cumulative Distribution Function (CDF) gives, for each value x, the probability that a random variable X takes a value less than or equal to x. Empirically, the CDF at value x is the fraction of observations that are at most x. The CDF is a non-decreasing function ranging from 0 to 1. Unlike the PMF (which can be noisy and sensitive to bin size), the CDF is smooth and shows the full shape of the distribution in a single curve. CDFs are useful for computing percentiles, comparing two distributions (a CDF shifted to the left means smaller values), and testing goodness-of-fit against theoretical distributions using the Kolmogorov-Smirnov test.

TOPIC 08

Percentile-Based Statistics and Random Numbers

The p-th percentile of a dataset is the value below which p% of the observations fall. The 25th, 50th, and 75th percentiles are quartiles (Q1, Q2/median, Q3). The interquartile range (IQR = Q3 - Q1) is a robust measure of spread not affected by extreme outliers. Percentile rank converts a raw score into its relative standing: a score at the 90th percentile is higher than 90% of all scores. In Python, np.percentile(data, 95) computes the 95th percentile. Random numbers are generated using pseudorandom number generators seeded with an initial value â€” np.random.seed(42) ensures reproducibility. Truly random variation should be distinguished from systematic bias, which shifts all measurements in the same direction.

TOPIC 09

Comparing Percentile Ranks Across Distributions

Comparing raw values across distributions is misleading when distributions have different means and spreads. For example, a score of 70 on a hard exam might be exceptional (90th percentile), while the same score on an easy exam might be below average (40th percentile). Percentile ranks normalize across distributions, enabling fair comparison. A complementary approach is standardization (z-scoring): z = (x - mean) / standard_deviation converts any value to the number of standard deviations above or below the mean, enabling comparison across datasets with different units and scales. Plotting CDFs or z-scored distributions side by side is a powerful way to visually compare two groups on a common scale.

TOPIC 10

Exponential, Normal, and Lognormal Distributions

The normal (Gaussian) distribution is the bell-shaped distribution described by its mean and standard deviation, and arises naturally whenever many independent random factors add together (Central Limit Theorem). The exponential distribution models the time between independent random events â€” for example, the time between customer arrivals â€” with the memoryless property (the probability of the next event does not depend on how long you have already waited). The lognormal distribution arises when the logarithm of the variable is normally distributed; it models quantities that are the product of many independent factors, such as stock prices, city populations, and biological measurements. Evaluating distributional fit uses Q-Q plots: if data follows the theoretical distribution, points align along a straight diagonal line.

Module notes: Statistical thinking focuses on understanding patterns in collections of values rather than interpreting isolated observations. Histograms show how values are distributed, while PMFs and CDFs provide probability-based descriptions of discrete and continuous data. Percentiles help compare observations relative to a distribution, and common probability models such as the normal and exponential distributions provide useful mathematical approximations for real-world variation.